Smart RTO - Hardware Connection Guide

Arduino Uno Prototype - 5V USB only, no mains. For CS students starting from zero.

1. Parts Required

- Arduino Uno + USB data cable (not charge-only)
- Breadboard (half or full size)
- Red, Yellow, Green LEDs x1 each (long leg = +, short leg = -)
- 220 ohm resistors x3 (red-red-brown-gold)
- IR obstacle sensor module (VCC, GND, OUT)
- 5V active buzzer (+ / -, long leg = +)
- Jumper wires 10-12 male-to-male (minimum workable: 7)
- Laptop with Arduino IDE + webcam

2. Breadboard Basics

Side rails marked + and - run the full length and are each one connected strip: use them for 5V and GND. In the middle, every group of 5 holes in a row is connected, but nothing connects across the centre gap. An LED sits across the gap: long leg one side, short leg the other. The resistor goes between the Arduino wire and the long leg.

3. Master Connection Table

Arduino	Component	Detail
Pin 8	Red LED +	Via 220 ohm; LED - to GND
Pin 9	Yellow LED +	Via 220 ohm; LED - to GND
Pin 10	Green LED +	Via 220 ohm; LED - to GND
Pin 2	IR OUT	VCC to 5V, GND to GND (LOW = obstacle)
Pin 3	Buzzer +	Buzzer - to GND (5V active, <30mA)
5V	+ rail	Feeds IR sensor VCC
GND	- rail	Common ground for all
USB	Laptop	Serial 9600 baud + power

4. Wiring Steps

Step 0 - Power rails (2 wires)

- Arduino GND -> breadboard - rail (black wire).
- Arduino 5V -> breadboard + rail (red wire).

Step 1 - Three LEDs (Pins 8, 9, 10)

- Place red LED across centre gap.
- Pin 8 -> resistor -> LED long leg.
- LED short leg -> GND rail.
- Repeat: Yellow on Pin 9, Green on Pin 10, same pattern.

Step 2 - IR sensor (Pin 2)

- VCC -> + rail (5V).
- GND -> - rail (GND).

- OUT -> Arduino Pin 2.
- Keep blue sensitivity knob in middle; 5-15 cm is enough.

Step 3 - Buzzer (Pin 3)

- Buzzer + -> Arduino Pin 3.
- Buzzer - -> GND rail.

5. If Wires Are Short

- Bend LED short legs directly into GND rail to save 3 wires.
- Plug resistor legs directly beside LED long legs.
- Minimum demo works with Red LED + IR + buzzer only (5-6 wires). Add Yellow/Green later; no code change needed.

6. Upload Code

- Open hardware/arduino/traffic_signal.ino in Arduino IDE.
- Select Board: Arduino Uno and correct Port.
- Press Upload. LEDs cycle: GREEN 8s -> YELLOW 2s -> RED 8s.
- Serial 9600 baud: SIGNAL:RED/YELLOW/GREEN + VIOLATION_TRIGGERED on RED+IR.

7. Standalone Test (no software)

- Watch full cycle: green 8s, yellow 2s, red 8s.
- During RED only, wave hand before IR: buzzer sounds 2s.
- GREEN/YELLOW + hand = nothing. That is correct stop-line logic.
- If never sounds, turn blue potentiometer slowly until ~10 cm range.

8. Connect to Dashboard

- Find port: ls /dev/ttyACM* /dev/ttyUSB* before/after plugging.
- In .env set ARDUINO_PORT=(your port) and DEMO_MODE=0.
- Run: source .venv/bin/activate, then python3 run.py.
- Open http://localhost:5000. Strip must show Arduino: CONNECTED.
- Trigger IR during RED: Live page shows violation with photo instantly.

9. Safety

- 5V USB prototype only. No mains, relays, motors, batteries.
- Every LED must have its 220 ohm resistor.
- If anything gets hot, unplug USB and check VCC/GND polarity.
- Port denied? sudo usermod -aG dialout $USER, then re-login.

10. Troubleshooting

Symptom	Cause	Fix
No LEDs	Code not uploaded	Re-upload; check Board+Port
One LED dark	LED reversed	Swap legs: long = +
Buzzer silent	IR wiring/sensitivity	Check OUT-Pin2; turn knob
Buzzer always on	IR stuck LOW	Clear sensor view
Port denied	dialout group	usermod + relogin

DISCONNECTED	Wrong port in .env	Fix ARDUINO_PORT, restart

Appendix - Serial Log

At 9600 baud expect: SIGNAL:GREEN ... SIGNAL:YELLOW ... SIGNAL:RED ... VIOLATION_TRIGGERED (only IR during RED). Python file backend/services/serial_listener.py parses exactly these lines.